

Mathematical Formulation of the Fish Metacom- munity Model

This document provides a detailed description of the spatially-explicit modeling framework used to simulate fish metacommunity dynamics in the Guadalquivir River basin. The model is implemented as a system of Ordinary Differential Equations (ODEs) that integrate local population dynamics, environmental filtering, and 3D dendritic dispersal.

1. General Model Structure

The rate of change for the population size $N_{i,s}$ of species s at site i is defined by the sum of local dynamics and net spatial flux:

$$\frac{dN_{i,s}}{dt} = \underbrace{f_{local}(N_{i,s}, \mathbf{N}_i, E_i)}_{\text{Local Dynamics}} + \underbrace{\sum_{j \in \text{neighbors}} (m_{ji}N_{j,s} - m_{ij}N_{i,s})}_{\text{Spatial Flux}}$$

where: - $N_{i,s}$ is the population density of species s at site i . - $\mathbf{N}_i$ is the vector of all species populations at site i . - E_i represents the local environmental conditions. - m_{ij} is the dispersal rate from site i to site j .

2. Local Dynamics and Environmental Filtering

The local component follows a Generalized Lotka-Volterra (GLV) structure, modified by environmental suitability filters.

2.1. Population Growth and Biotic Interactions

The local dynamics are expressed as:

$$f_{local} = N_{i,s} \left(r_{i,s}^{eff} - \sum_{j=1}^S \alpha_{sj} N_{i,j} \right)$$

where: - $r_{i,s}^{eff}$ is the effective intrinsic growth rate. - α_{sj} is the interaction coefficient representing the effect of species j on species s (asymmetric interaction matrix).

2.2. Environmental Filtering

The effective growth rate $r_{i,s}^{eff}$ is determined by the base growth rate r_s and a site-specific suitability factor $W_{i,s}$:

$$r_{i,s}^{eff} = r_s \cdot W_{i,s}$$

The suitability factor $W_{i,s}$ incorporates thermal tolerance and habitat quality:

$$W_{i,s} = \text{ThermalFilter}(T_i, \text{opt}_s, \sigma_s) \cdot h_i$$

Thermal Filter We use a Gaussian-shaped “thermal window” to model species-specific temperature tolerances:

$$\text{ThermalFilter}(T_i, \text{opt}_s, \sigma_s) = \exp\left(-\frac{(T_i - \text{opt}_s)^2}{2\sigma_s^2}\right)$$

where: - T_i is the temperature at site i . - opt_s is the optimal temperature for species s . - σ_s is the thermal tolerance (standard deviation) of species s .

Habitat Suitability

- h_i is the habitat suitability index at site i , which accounts for anthropogenic stressors (e.g., conductivity), river width, and mesohabitat types.

3. Spatial Flux (3D Dendritic Dispersal)

The spatial component models the movement of individuals through the river network, accounting for hydrological distance, elevation changes, and physical barriers.

3.1. Dispersal Rate Formulation

The dispersal rate m_{ij} from site i to site j is defined as:

$$m_{ij} = m \cdot \left(\frac{1}{d_{ij}}\right) \cdot x_{ij} \cdot p_{ij}$$

where: - m is the base dispersal intensity. - d_{ij} is the hydrological distance between site i and j . - x_{ij} is the 3D connectivity factor (elevation cost). - p_{ij} is the dam passability factor (ranging from 0.0 to 1.0).

3.2. Elevation Cost (Upstream vs. Downstream)

The cost of movement depends on the relative elevation of the sites:

- **Downstream or Equal Elevation** ($e_j \leq e_i$):

$$x_{ij} = 1.0$$

- **Upstream** ($e_j > e_i$):

$$x_{ij} = \frac{1}{1 + c(e_j - e_i)}$$

where: - e_i and e_j are the elevations of the source and destination sites, respectively. - c is the upstream migration cost constant.

4. Implementation Details

- **State Vector:** The system state u is a matrix of size $N_{sites} \times N_{species}$.
- **Sparse Matrix Optimization:** To handle large-scale river networks efficiently, the dispersal rates are pre-calculated into a sparse matrix M , where M_{ji} represents the rate from i to j .
- **Numerical Integration:** Given the potential stiffness arising from the coupling of fast dispersal and slower local dynamics, the system is solved using stiff-aware solvers (e.g., Rodas5 or TRBDF2).

5. Model Assumptions

1. **Constant Interaction Matrix:** Biotic interaction coefficients (α_{sj}) are assumed to be constant across the entire basin, though the realized interaction depends on local densities.
2. **Passive/Active Dispersal Balance:** Dispersal is modeled as a density-dependent flux proportional to the population at the source site.
3. **Gaussian Thermal Niche:** Species performance is assumed to decay symmetrically as temperatures deviate from the optimum.
4. **Dam Passability:** Dams act as semi-permeable filters that reduce the effective dispersal rate between connected nodes.